

- How much water does the overhead tank in your house can hold?
- Does your friend have fever? What is his body temperature?

You have learnt about many types of numbers so far. Now is the time to extend the ideas further.

2.2 Rational Numbers

When you want to count the number of books in your cupboard, you start with 1, 2, 3, … and so on. These counting numbers 1, 2, 3, … , are called Natural numbers. You know to show these numbers on a line (see Fig. 2.1).



Fig. 2.1

We use $\mathbb{N}$ to denote the set of all **Natural numbers**.

$$\mathbb{N} = \{ 1, 2, 3, \ldots \}$$

If the cupboard is empty (since no books are there). To denote such a situation we use the symbol 0. Including zero as a digit you can now consider the numbers 0, 1, 2, 3, … and call them **Whole numbers**. With this additional entity, the number line will look as shown below



Fig. 2.2

We use $\mathbb{W}$ to denote the set of all Whole numbers.

$$\mathbb{W} = \{ 0, 1, 2, 3, \ldots \}$$

Certain conventions lead to more varieties of numbers. Let us agree that certain conventions may be thought of as **"positive"** denoted by a '+' sign. A thing that is 'up' or 'forward' or 'more' or 'increasing' is positive; and anything that is 'down' or 'backward' or 'less' or 'decreasing' is **"negative"** denoted by a '–' sign.

You can treat natural numbers as positive numbers and rename them as **positive integers**; thereby you have enabled the entry of negative integers –1, –2, –3, … .

Note that –2 is "more negative" than –1. Therefore, among –1 and –2, you find that –2 is smaller and –1 is bigger. Are –2 and –1 smaller or greater than –3? Think about it.

The number line at this stage may be given as follows:



Fig. 2.3

2-Real Numbers.indd 42 02/12/2022 21:32:44